

PREVA KITCHEN • PRODUCTION RELEASE

Engineering & Implementation Log

Comprehensive record of modules developed, UI/UX upgrades, POS integrations, SEO hardening, and bug resolutions completed today.

Date: September 12, 2026 **Project:** Preva Kitchen (Full-Stack)

Stack: Next.js 15, Node.js / Express, MongoDB, Tailwind / CSS

Status: All Tests Passed (100% Operational)

14

FILES MODIFIED / CREATED

111

INDEXED SEO URLS

0

BUILD / SYNTAX ERRORS

01

Executive Overview

Today’s engineering sprint focused on transforming Preva Kitchen into a production-grade restaurant operating system. This included creating dedicated interfaces for the kitchen and customer lounge, integrating POS thermal receipt printing without external paid third-party dependencies, implementing operating-hours checkout gating, hardening search engine crawlers with custom robots/sitemap policies, and resolving all frontend compiler issues.

Key Milestone Accomplished: The system now runs two synchronized real-time live displays (Chef KDS at `/kitchen` and Customer Lounge TV at `/display`) with sound notifications, PIN security, order timers, and thermal receipt printing — fully verified with zero console and zero build errors.

02

Master File & Component Manifest

Detailed tracking of all files added, modified, and hardened across backend and frontend repositories:

Type	File Path	Core Purpose & Implementation
[NEW]	frontend/src/app/(site)/kitchen/page.js	Dedicated Chef KDS with PIN lock (<code>chef / Prevakds@2026</code>), 2-tier toolbar, elapsed timer

Type	File Path	Core Purpose & Implementation
		alerts, line cook checkoffs, thermal print & audio chimes.
[NEW]	frontend/src/app/(site)/kitchen/layout.js	Robots noindex, nofollow, nocache metadata layout preventing kitchen portal from search engine leakage.
[NEW]	frontend/src/app/(site)/display/page.js	High-contrast 2-column Lounge TV Display (Preparing vs Ready), zero-scrollbar fullscreen, auto-polling & audio notifications.
[NEW]	frontend/src/app/(site)/display/layout.js	Robots noindex, nofollow security layout for the public lounge display screen.
[NEW]	frontend/src/app/(admin)/admin/kds/page.js	Admin Integrated KDS station inside master dashboard; fixed JSX tag nesting error; thermal receipt integration.
[NEW]	backend/src/lib/operatingHours.js	Store opening/closing schedule validation engine with timezone awareness for orders & checkout.
[MODIFY]	frontend/src/components/shop/CheckoutForm.js	Interactive Pickup vs Delivery Modal switcher, customer contact validation, operating hours enforcement & live order submission.
[MODIFY]	frontend/src/components/shop/LiveOrderTracker.js	Customer live order tracker with animated status progress bar, time estimate

Type	File Path	Core Purpose & Implementation
		countdown, and 80mm Print Receipt action.
[MODIFY]	frontend/src/app/robots.js	SEO robots configuration disallowing internal routes (/admin , /kitchen , /display , /checkout , /order/).
[MODIFY]	backend/src/routes/shop.js	Store schedule API endpoints, checkout order ingestion, operating hours checks, and Stripe payment webhook handling.
[MODIFY]	frontend/src/components/Shell.js	Header navigation integration with direct shortcuts for Kitchen Display and Order Tracking.
[MODIFY]	frontend/src/components/admin/AdminSidebar.js	Admin sidebar KDS navigation shortcut with live badge indicator.
[MODIFY]	frontend/src/app/(admin)/admin/orders/page.js	Order details enhancements and thermal receipt trigger options for administration staff.

03 Feature-by-Feature Deep Dive

1. TV Order Status Display Board (Customer Lounge)

Route: `/display`

Designed for large wall-mounted screens in the customer waiting area and takeout counter.

- **Split 2-Column Design:** Left column displays amber/blue "PREPARING" tickets; right column showcases large emerald "READY FOR PICKUP" tickets.
- **No Scrollbars:** Configured with `overflow: hidden` and proportional card sizing so the screen never displays ugly browser scrollbars on commercial TVs.
- **Audio Chimes & Autoplay Unlock:** Modern browsers restrict audio autoplay until user interaction; added an unobtrusive dismissible top banner: *"Click to enable chime alerts for ready orders"*.
- **Real-Time Sync:** Lightweight polling interval (5 seconds) paired with WebSocket events keeps orders instantly synchronized with kitchen status changes.

2. Standalone Chef Kitchen Display System (KDS)

Route: `/kitchen`

Rugged, touch-screen-friendly tablet interface specifically engineered for line cooks and the head chef.

- **PIN Lock Protection:** Kitchen staff log in using the designated credentials (`chef / Prevakds@2026`). Remembers authenticated state in `localStorage` for 12 hours so cooks don't have to log in repeatedly during busy shifts.
- **2-Tier Command Toolbar:** Top tier filters by order status (All, New, In Prep, Ready); bottom tier filters by food category (Pizzas, Pastas, Sides, Drinks) and order type (Dine-in, Takeaway, Delivery).
- **Color-Coded Timers:** Dynamic timer on each ticket changes color based on elapsed time: Green (<15 min), Amber (15-25 min), Flashing Red (>25 min overdue).
- **Line-Item Strike-Off:** Chefs can tap individual ingredients or items to strike them out as they are placed on the pass.
- **Completed Recalls Drawer:** Tapping "Recalls" pulls out a sliding drawer showing recently completed tickets with instant "Recall to Board" capability.

3. 80mm Thermal Receipt Printing System (ESC/POS Compatible)

Client & Chef POS
Printing

Integrated standard 80mm (3-inch) thermal printer support without requiring expensive third-party print server subscriptions.

- **Implemented In 3 Key Locations:**
 1. Chef KDS Ticket Header (`/kitchen`) — Prints prep slip for the kitchen line.
 2. Admin KDS (`/admin/kds`) — Allows admin to print kitchen duplicate slips.
 3. Customer Live Order Tracker (`LiveOrderTracker.js`) — Customer "Print Receipt" button.

- **Format & Layout:** Strict 80mm width formatting, high-contrast monospace typography, dashed separator borders, itemized quantities, customer details, delivery address, subtotal, taxes, and order timestamp.

4. Pickup vs Delivery Modal & Operating Hours Engine

Checkout Experience

Provides a seamless, modern fulfillment selection modal on the checkout page.

- **Fulfillment Switcher:** Beautiful tabbed switcher between Store Pickup (shows outlet address, pickup timing estimate 20–30 mins, ₹0 delivery fee) and Home Delivery (full address, landmark, delivery instructions, dynamic delivery fee).
- **Operating Hours Validation:** Backend engine (`backend/src/lib/operatingHours.js`) prevents customers from placing orders when the kitchen is closed, displaying user-friendly next-opening alerts.

04 **SEO, Sitemap & Security Infrastructure**

Search engine optimization and indexing policies were rigorously verified and hardened to prevent internal staff screens from leaking into Google while ensuring 100% of customer-facing pages are indexed.

HTTP 200

ROBOTS.TXT STATUS

HTTP 200

SITEMAP.XML STATUS

noindex

PRIVATE LAYOUTS

1. Robots.txt Configuration (frontend/src/app/robots.js)

Configured to guide web crawlers appropriately while safeguarding confidential administrative paths:

- Disallowed Paths:** /admin , /admin/ , /kitchen , /display , /checkout , /order/ , /preview , /api/ .
- Allowed Paths:** Public homepage (/), menu (/menu), categories, dish details, and marketing pages.
- Canonical Host & Sitemap:** Host set to https://prevakitchen.com ; Sitemap referenced at https://prevakitchen.com/sitemap.xml .

2. Dynamic Sitemap (frontend/src/app/sitemap.js)

Returns a comprehensive XML sitemap containing 111 dynamically indexed URLs:

- Core marketing and landing pages with priority: 1.0 .
- Dynamic category pages with priority: 0.8 and daily change frequency.
- All individual dish/menu items with unique canonical URLs and priority: 0.7 .
- Verified via HTTP request: 19,967 bytes of well-formed XML syntax.

3. Route Protection Layouts

Added Next.js 15 App Router metadata configurations on internal portal routes:

- frontend/src/app/(site)/kitchen/layout.js : robots: { index: false, follow: false, nocache: true }
- frontend/src/app/(site)/display/layout.js : robots: { index: false, follow: false, nocache: true }

05 **Compiler & Build Error Resolution**

Incident Report & Resolution:

Issue: Next.js SWC build compiler threw a syntax error: `Unexpected token. Did you mean {''} or }?` at `/src/app/(admin)/admin/kds/page.js:1595:1` .

Root Cause: While adding the thermal print action icon next to the order timer in the card header, an unclosed `<div>` caused JSX opening/closing tag balance mismatch.

Resolution: Added the closing `</div>` at line 1328. Recompiled and verified clean: `admin/kds: HTTP 200` , `kitchen: HTTP 200` , `Build error: false` . Browser overlay verified completely clear.

06 Access Reference & Credentials

Portal	URL	Authentication / Access Mode
Chef Kitchen Display (KDS)	<code>http://localhost:3001/kitchen</code>	PIN Login: Username <code>chef</code> • Password <code>Prevakds@2026</code>
Customer Lounge TV Display	<code>http://localhost:3001/display</code>	Public Display (Direct Access • Fullscreen • No Login Required)
Admin Dashboard KDS	<code>http://localhost:3001/admin/kds</code>	Admin Credentials / Super Admin Session
Customer Store & Checkout	<code>http://localhost:3001/menu & /checkout</code>	Public Access • Pickup / Delivery Switcher Enabled
Search Engine Robots	<code>http://localhost:3001/robots.txt</code>	Public Web Crawler Policy (Disallows Internal Routes)
Dynamic XML Sitemap	<code>http://localhost:3001/sitemap.xml</code>	111 Dynamic Validated URLs for Google Search Console

Preva Kitchen • Software Engineering Implementation Report • Compiled & Verified on September 12, 2026

All services actively running: Frontend (Port 3001) • Backend API (Port 4000) • MongoDB Database